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REQUEST FOR INFORMATION (RFI) – No. 42426-RD [Mid-energy electron beam (eBeam) systems]

1.0 - Background Information

RFI Release date	**April 24, 2026**
Questions deadline	**May 18, 2026**
RFI Response due date	**June 1, 2026**
Submit to:	**Randy Dykes**
Phone Number	**509-375-2023**
Email Address	**randy.dykes@pnnl.gov**

- ✓ **Battelle Memorial Institute (Battelle), Management and Operations Contractor of the U.S. Department of Energy's (DOE) Pacific Northwest National Laboratory (PNNL) in Richland, Washington, requests information as detailed below:**

- ✓ **This RFI form is designed to collect information from interested Offerors for consideration by PNNL. All responses should be accurate, complete, and submitted by the deadline specified in the solicitation.**

- ✓ **Confidentiality: Proprietary information submitted as part of the response to the RFI must be clearly identified and marked as PROPRIETARY (*if applicable*).**

- ✓ **This RFI <u>does not constitute a commitment to contract or purchase</u>. Responses to this RFI are for informational purposes only and will be used to inform future procurement decisions.**

1.1 - Response Format

- ✓ Preferred format: searchable PDF (plus Excel for pricing tables and spares lists if desired).
- ✓ Limit narrative response to 25 pages excluding datasheets/appendices.
- ✓ Include a requirements compliance matrix (Requirement / Comply Y/N / Notes / Exceptions).

1.2 - Purpose

- The Pacific Northwest National Laboratory (PNNL) is seeking information from qualified suppliers regarding mid-energy electron beam (eBeam) systems intended to support medical device sterilization applications.

- The goal of this RFI is to compile available technical approaches, configurations, safety/shielding concepts, footprint/integration constraints, validation-support capabilities, likely lead times, etc., to craft the appropriate RFP that will solicit proposals with final prices for a self-shielded system deployable at end-of-line (EOL) with optional in-line capability for terminal and adjunctive sterilization of medical products. Shuttle systems will be considered, but the desired configuration includes continuous conveyance that supports transport of product through the beam zone directly on conveyor or in standard trays on conveyor.

1.3 - Offeror Information

Company Name	
Headquarters / Manufacturing Locations, and industries served	
Mailing Address	
Point of Contact Name	
Title/Position	
Phone Number	
Email Address	

1.4 - Company Qualifications

Years delivering eBeam systems in the 1–2 or 2-3 MeV range	
Number of comparable 2 MeV systems delivered/installed; example customers (if permitted).	

Quality program overview (e.g., ISO certifications), service organization coverage.	
Small Business Status (Yes/No; If Yes, specify type)	

2.0 - Background / Use Case

The desired system will be used to irradiate low- and medium- density (0.08 to 0.2 g/cm^3) products in their primary packaging (not boxed or in shippers). The system is envisioned to enable evaluation, acceptance testing, and eventual adoption by industry stakeholders. Respondents to the anticipated future RFP will describe how their proposed solution supports sterilization-validation activities and operational needs in regulated environments. Respondents to this RFI will describe any compliance issues with each requirement below and, if possible, will describe some details of what system solutions could be offered.

3.0 - Draft of High-Level Technical Requirements (Minimum/Target)

The items below are presented in this RFI to allow comments and refinement of requirements for the anticipated RFP that is being written and will be later posted. In response to this RFI, it is not necessary to submit detailed responses to the technical requirements below. Rather it is requested that vendors clearly identify any exceptions necessary to the items below, while providing clarifications and concerns about the specifications that are expected in the anticipated RFP.

3.1 eBeam / Accelerator

Energy: 2 MeV (vendor should indicate full energy range, over which (if any) the eBeam is able to easily and smoothly adjust energy to other values that can be used routinely)

Beam Power: 2–10 kW (vendor provide ranges that could be available).

Vacuum Pumps: Provide summary of vacuum system.

Scan Horn: Standard down-fire beam with 1 m-wide scan horn (describe beam width at product plane (passing horizontally under vertically down-firing beam), beam uniformity at product plane, scan frequency, and beam percentage utilization). If width of less than 1 m is preferred by vendor, available widths should be indicated. If scan width is operator-adjustable, vendor should specify range of adjustability and whether narrower beam widths require lowering the maximum beam power that can be used. Describe cooling provided for window foil.

Product-release by Dose Control: Describe tolerance windows recommended to be around energy stability, beam current stability, scan control, conveyor speed stability, and any other parameter affecting dose delivery and its cumulative uncertainty. Describe monitoring methods for recording all critical parameters affecting dose delivery. Describe warning processes and responses to a parameter value occurring outside its tolerance window (and therefore jeopardizing sterility assurance claims, which are mandatory for product release).

3.2 System Configuration

Self-shielded cabinet/system design suitable for industrial deployment in unrestricted areas (perimeter dose below 0.5 µSv/hour).

Primary Configuration: End-of-line (EOL) or Off-line integration.

Optional Capability: In-line integration (describe what dedicated space is required to enable in-line use, and any considerations for integration of controls to line controls).

Product Focus: Optimized for single-sided exposure of low-density products, e.g., 0.8 g/cm^2 (i.e., 4 cm of 0.2 g/cm^3, or 8 cm of 0.1 g/cm^3). Describe penetration expectations at 2 MeV and typical maximum product thicknesses for stated densities, if DUR ≤1.5.

3.3 Integration / Footprint

- Size/weight constraints for final integration: Vendor shall provide **in the RFP**:
- Overall dimensions (L×W×H) for assembled system, including shielding, under-beam conveyor, and ancillary equipment
- Total system weight and floor-loading requirements, not including conveyor infeed/outfeed and any ancillary components/cabinets
- Required space to allow clearances for service access to all parts of system
- Recommended layout drawings (top/side views)
- Conveyor height-range allowed (product dimensions provided below)
- Utilities required: electrical, cooling, compressed air, exhaust/ventilation, catalytic ozone destruction option network connections, for full-power operation 7x24.
- Describe quantity of transformer oil and/or sulfur hexafluoride (SF_6) gas (if any) that will be required for insulating high voltage components.

3.4 Safety, Shielding, and Compliance

- Describe shielding approach to achieve self-shielded operation in unrestricted area at the specified energy/power (more details requested in section 4.4).
- Preferred shielding design should include continuous conveyance with products directly on conveyor or in standard trays on conveyor. Shuttle system will also be considered but is not preferred.
- Describe standard interlocks, radiation monitoring, fail-safe behavior, risk-assessment method, Safety Integrity Level (SIL), access-controls, emergency stops, and if any video-monitoring of processing zone can be provided.
- Identify applicable standards/certifications supported (e.g., ISO/ASTM practices relevant to radiation processing equipment and dosimetry; any safety certifications typically provided).
- Identify electromagnetic compatibility (EMC) standards that are satisfied (emissions and immunity).

3.5 Throughput / Material Handling (Provide Ranges)

- Conveyor type(s) that vendor can supply, speed range, product presentation options (single pass, multi-pass, flip/turnover if available). Primary requirement is to convey a rectangular and uniform-density product (or a tray carrying products) with dimensions: 45 cm x 45 cm x 5 cm. For shielding design of labyrinth turns, assume minimum of 5 cm between sequential identical products or trays.
- Throughput capacities for above product in tray, requiring a 15 kGy minimum dose in product, for both a 0.15 g/cm^3 and a 0.2 g/cm^3 homogeneous-density case.
- Beam uniformity variations, as measured at top of the product plane and as absorbed by product moving through the beam.
- List maximum and minimum speeds allowed (for product trays shown above) through process zone.
- Provide typical start-up time required to reach full power for:

1) cold start (from being in typical weekend condition for two days and then re-started Monday morning)
2) warm start (from being left in the typical idle mode for 1 hr. during day and then re-started)
3) after short power interruption to system (0-10 min)
4) after complete power loss to system for 4 hours

4. Information Requested from Vendors for this RFI

Please respond if any information requested in the section below would be objectionable if it appeared as-is in the RFP. If a section is believed not applicable, state “N/A” with explanation. In the sections below marked with **“For RFI”**, provide complete responses as much as possible.

4.1 Technical Solution Description

- Proposed accelerator technology and architecture (e.g., type, scanning method), energy spectrum at vacuum-side of window foil at 2 MeV.
- Uptime-performance capabilities, if operating 7x24 at full power at 2 MeV and 2–10 kW (include datasheets, if available).
- Scan-horn details (confirm 1 m beam width, at product plane; provide preferred height for window above conveyor; window area dimensions).
- Width of uniform beam at product plane, 5 cm above conveyor surface
- Is beam-monitoring technology available to confirm the beam current and energy as it exits the window?
- Expected surface dose rate (i.e., dose x tray speed, kGy m/min) at 2 MeV and maximum power, when delivering minimum of 15 kGy dose into 0.1 g/cm^3 product given above operating envelope (other allowed energies and powers for machine; provide assumptions and indicate energy range below 2 MeV, if able to easily and smoothly adjust energy to routinely operate at other lower values).

4.2 Product/Density Application Fit

- Describe system efficiency (ratio of “kW of beam power reaching the product” to the “kVA coming from wall-power for full system”) for processing 0.1 g/cm^3 products at maximum throughput at 2 MeV, with 15 kGy minimum dose.
- Provide examples of products previously handled (dimensions, densities, minimum doses, throughputs).
- Any recommended fixturing/packaging considerations for the expected throughputs, speeds and handling.

4.3 Self-Shielding and Facility Interface

- Shielding materials used and their maximum thicknesses for 2 MeV self-shielded system, and sample radiation dose-maps, if available, from previous similar systems for verification of approach.

Expected external dose rates at 10 cm from surfaces/slots during operation in unrestricted areas. Explain radiation-monitoring approach, if any can be provided.
- Facility requirements: electrical power drop required (kVA), cooling capacity required (kW, flow, inlet temperatures to eBeam machine), exhaust flow required to outside (in order to maintain negative pressure in product path through process zone), ambient environment restrictions (temperature and humidity limits), options for anchoring to floor, required floor-loading capacity under shield, required clearance for largest system crate or component to be brought into installation area.

4.4 Controls, Data, and Cyber/Connectivity

- Human-machine interface / programmable logic controller (HMI/PLC) platform, alarm/interlock architecture.
- Data logging (all dose-relevant critical parameters; run-record reports available), audit trails, user-access control.
- Interfaces: Ethernet/Industrial Protocol (IP), Open Platform Communications Unified Architecture (OPC UA), Modbus, etc.
- Remote-support-connectivity solution and cybersecurity considerations.

4.5 Testing Support and Validation

- Factory Acceptance Test (FAT), being performed at Supplier facility. List typical acceptance criteria, include power-efficiency tests, as well as demonstration of variations/control/recording of all dose-relevant critical eBeam system parameters affecting sterilization.
- Site Acceptance Test (SAT), being performed at customer site. List typical acceptance criteria.
- Commissioning approach and support through Installation Qualification (IQ) and Operational Qualification (OQ).
- Dosimetry process support for measurements on eBeam system (recommended procedures, holders, beam control of critical parameters for characterizing impact on dose).
- Documentation packages typically provided to assist process validation for sterilization.
- Support for regulatory engagement through Process Qualification (PQ).

4.6 Project Execution and Schedule

- Typical lead time from contract to FAT.
- Typical time from FAT completion to SAT at customer site in USA.
- Major project phases, including required (if any) phase-gate approvals and customer responsibilities.

- Training provided (operator, maintenance, radiation safety, process validation), along with durations, number of persons allowed, and preferred locations for each type of training.
- Warranty period, starting point, and terms.

4.7 Service and Sustainment

- Recommended preventive maintenance requirements (describing whether each operation is recommended to be done by end-user or vendor) and intervals.
- For the following three lists of spare parts, include:
  - their expected lifetimes
  - estimated lead times
  - prices
  - which are commercially available off-the-shelf versus proprietary to vendor.

1. List of standard spare parts that are included with system
2. List of critical spares to keep on site to maintain high uptime (whether also in the list of included standard spares or not).
3. List of additional recommended spares.

- Field service response times by phone and in person, during warranty and during continuing service contract.
- Service contract options. Whether 3rd party service providers may be used without voiding warranty.

4.8 **"For RFI"** – rough order of magnitude (ROM) pricing (final pricing provided at time of RFP). Provide budgetary estimates (ranges acceptable) broken out as applicable:

- Base system (accelerator, scan horn, self-shielding, standard conveyor for product in section 3.5, system controls, data-logging, cooling system, exhaust system).
- Shipping – No estimate required, as it will be dependent on final location(s). Please confirm if all components, when crated, will fit in Standard or High-Cube containers.
- Installation/commissioning at USA site.
- Training of operators and maintenance personnel.
- Optional in-line integration features.
- Optional product-handling features.
- Annual service/support costs after warranty period.

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4.9 Risks and Constraints

- Key technical or operational risks for meeting the stated requirements.
- Mitigation strategies (e.g., redundancies, design alternatives).
- Any assumptions required to meet performance and validate system for sterilization.

5. **"For RFI"** - Requested Attachments (if available) (and in future for RFP)

- Product brochures/datasheets for 1–2 MeV systems.
- Example layout drawings and utility requirements.
- Sample FAT/SAT or acceptance test plan outline.
- List of applicable standards supported.
- Example radiation survey/verification documentation for self-shielded units in unrestricted areas in this energy range, if available.

~End of RFI~